Convex Optimization: Problem Set 1

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Some problems borrowed from Optimization course of Daniel Dadush.

1. Recall the definition of convex hull:

$$\text{conv}(S) := \left\{ \sum_{i=1}^{N} \lambda_i \mathbf{x}_i \mid \mathbf{x}_i \in S, \lambda \geq 0, \sum_{i=1}^{N} \lambda_i = 1 \right\}.$$

   1. Prove that for finite $S, \text{conv}(S) = \cap_{K \supseteq S} K$ where the intersection is over all closed convex K containing S. Therefore $\text{conv}(S)$ is the smallest convex set containing S.
   2. Prove Jensen's inequality: for convex $f : \mathbb{R}^n \to \mathbb{R}$ and input $\mathbf{x} = \sum_{i=1}^{N} \lambda_i \mathbf{x}_i$ where $\lambda_i \geq 0, \sum_{i=1}^{N} \lambda_i = 1$

$$f(\mathbf{x}) \leq \sum_{i=1}^{N} \lambda_i f(\mathbf{x}_i).$$

   3. Show that if $\mathcal{C} = \text{conv}\{x_1, ..., x_m\} \subseteq \mathbb{R}^n$ is a compact convex set and $f : \mathbb{R}^n \to \mathbb{R}$ is a convex continuous function then the **supremum** of f over $\mathcal{C}$ is attained at one of the vertices x_j.

Solution:

1. Clearly any convex $K \supseteq S$ contains any convex combination $\sum_{i=1}^{N} \lambda_i \mathbf{x}_i$ where $\mathbf{x}_i \in S$, so $\text{conv}(S) \subseteq \cap_{K \supseteq S} K$. Conversely, if $x \notin \text{conv}(S)$ then there exists a separating hyperplane $\langle w, x \rangle > \nu := \sup_{y \in \text{conv}(S)} \langle w, y \rangle$ and so the halfspace

$$H := \langle y \mid \langle y, w \rangle \leq \nu \}$$

   is a convex set satisfying $H \supseteq S, y \notin H$.

2. We will prove the statement by induction on the number of points N. Notice that the case $N = 2$ is exactly the definition of convexity for f. Now for

$N \geq 2$, given $N+1$ points $\mathbf{x}_1, ..., \mathbf{x}_{N+1}$ and non-zero coefficients $\lambda \in \mathbb{R}^{N+1}_{++}$ with $\sum_{i=1}^{N+1} \lambda_i = 1$, let $\tilde{\lambda} := \sum_{j=1}^{N} \lambda_j = 1 - \lambda_{N+1}$ and

$$\mathbf{x} := \sum_{i=1}^{N} \frac{\lambda_i}{\tilde{\lambda}} \mathbf{x}_i.$$

This is a convex combination of $\mathbf{x}_1, ..., \mathbf{x}_N$ as $\gamma_i := \frac{\lambda_i}{\tilde{\lambda}}$ satisfies $\gamma \in \mathbb{R}^N_+$ and $\sum_{i=1}^{N} \gamma_i = 1$. Therefore we can bound the function as

$$\begin{aligned} f\big(\sum_{i=1}^{k+1} \lambda_i \mathbf{x}_i\big) &= f(\tilde{\lambda}\mathbf{x} + (1-\tilde{\lambda})\mathbf{x}_{N+1}) \leq \tilde{\lambda} f(\mathbf{x}) + (1-\tilde{\lambda}) f(\mathbf{x}_{N+1}) \\ &= \tilde{\lambda} f\big(\sum_{i=1}^{N} \frac{\lambda_i}{\tilde{\lambda}} \mathbf{x}_i\big) + \lambda_{N+1} f(\mathbf{x}_{N+1}) \leq \sum_{i=1}^{N} \lambda_i f(\mathbf{x}_i) + \lambda_{N+1} f(\mathbf{x}_{N+1}), \end{aligned}$$

where in the first step we used the definitions of $\mathbf{x}, \tilde{\lambda}$, in the second step we used the base case, and in the final step we used the induction hypothesis.

3. By continuity of f and compactness of $\mathcal{C}$, $x^* := \arg\max_{x \in \mathcal{C}} f(x)$ is attained at some point. Now let $x^* = \sum_{i=1}^{N} \lambda_i x_i$ where $\lambda \geq 0, \sum_{i=1}^{N} \lambda_i = 1$. Then by Jensen's inequality we can bound

$$f(x^*) \leq \sum_{i=1}^{N} \lambda_i f(x_i) \leq \max_{i \in [N]} f(x_i),$$

where the last step is because λ is a convex combination. Since x^* is a maximizer, the statement is shown.

2. Show the following are equivalent and prove any one of them:

   (a) Cauchy-Schwarz inequality for $\mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^{n} x_i y_i \leq \sqrt{\sum_{i=1}^{n} x_i^2} \sqrt{\sum_{i=1}^{n} y_i^2} = \|x\|_2 \|y\|_2;$$

   (b) The triangle inequality $\|x + y\|_2 \leq \|x\|_2 + \|y\|_2$;

   (c) Convexity of the Euclidean norm $f(x) := \|x\|_2$;

   (d) Convexity of the unit ball $B_2^n := \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\}$.

Solution: We first show equivalence of (a) Cauchy-Schwarz implies (b) the triangle inequality: first assuming (a), we have

$$\|x+y\|_2^2 = \|x\|_2^2 + 2\langle x, y\rangle + \|y\|_2^2 \leq \|x\|_2^2 + 2\|x\|_2\|y\|_2 + \|y\|_2^2 = (\|x\|_2 + \|y\|_2)^2,$$

where the first step was by definition $\|z\|_2^2 = \langle z, z\rangle$, and in the second step we used Cauchy-Schwarz.

For the converse, we have

$$0 \leq (\|x\|_2 + \|y\|_2)^2 - \|x+y\|_2^2 = 2\|x\|_2\|y\|_2 - 2\langle x, y\rangle,$$

where the first inequality is by the triangle inequality. The result follows by rearranging.

Next, we note that (b) the triangle inequality is equivalent to (c) convexity of $f(x) = \|x\|_2$ by homogeneity.

Next, we show (c) the triangle inequality implies (d) convexity of B_2^n: consider $x, y \in B_2^n$ and $\lambda \in [0, 1]$, then

$$\|(1-\lambda)x + \lambda y\|_2 \leq (1-\lambda)\|x\|_2 + \lambda\|y\|_2 \leq (1-\lambda) + \lambda = 1,$$

where the first step is by triangle inequality and homogeneity of the Euclidean norm, and in the last steps we used the assumption that $\|x\|_2, \|y\|_2 \leq 1$; therefore the convex combination is also in B_2^n.

For the converse, consider any $x, y \in \mathbb{R}^n$ and note $\frac{x}{\|x\|_2}, \frac{y}{\|y\|_2} \in B_2^n$ (assuming neither is 0 as this case is trivial). Therefore we have that any convex combination of these is in B_2^n, so in particular

$$\frac{x+y}{\|x\|_2 + \|y\|_2} = \frac{\|x\|_2}{\|x\|_2 + \|y\|_2}\frac{x}{\|x\|_2} + \frac{\|y\|_2}{\|x\|_2 + \|y\|_2}\frac{y}{\|y\|_2} \in B_2^n,$$

where we used the convex combination with $\lambda := \frac{\|y\|_2}{\|x\|_2 + \|y\|_2}$. By definition of B_2^n this implies

$$\left\| \frac{x+y}{\|x\|_2 + \|y\|_2} \right\|_2 \leq 1 \quad \iff \quad \|x+y\|_2 \leq \|x\|_2 + \|y\|_2,$$

where we used homogeneity of the Euclidean norm.

Finally, we prove the Cauchy-Schwarz inequality. We apply the elementary fact $(a+b)^2 \geq 0 \implies 2ab \leq a^2 + b^2$:

$$2\sum_{i=1}^n x_i y_i \leq \min_{\lambda > 0} \sum_{i=1}^n \lambda x_i^2 + \lambda^{-1} y_i^2 = \min_{\lambda > 0} \lambda\|x\|_2^2 + \lambda^{-1}\|y\|_2^2 \leq 2\|x\|_2\|y\|_2,$$

where the first step was by applying $2ab \leq a^2 + b^2$ for $a = \sqrt{\lambda}x_i, b = y_i/\sqrt{\lambda}$, and in the last step we chose $\lambda := \|y\|_2/\|x\|_2$.

3. For convex closed $f : \mathbb{R}^n \to \mathbb{R}$ and $x \in \mathrm{dom}(f)$:

   - Show the sub-level set $L := \{y \mid f(y) \leq f(x)\}$ is a convex set.
   - Show for any subgradient $g \in \partial f(x)$, $(g, -1)$ gives a *supporting hyperplane* at $(x, f(x))$ for the epigraph

   $$\mathrm{epi}(f) := \{(x, t) \mid f(x) \leq t\}.$$

   - Show that $g \in \partial f(x)$ gives a *supporting hyperplane* at x for the sub-level set

   $$L := \{y \in \mathbb{R}^n \mid f(y) \leq f(x)\}.$$

   Solution:

   - Consider $y, z \in L$ and $\lambda \in [0, 1]$, then

   $$f((1-\lambda)y + \lambda z) \leq (1-\lambda)f(y) + \lambda f(z) \leq (1-\lambda)f(x) + \lambda f(x) = f(x),$$

   where the first step is by convexity, and in the second we used that $y, z \in L$ are in the sub-level set. This shows $(1-\lambda)y + \lambda z \in L$, establishing convexity.

   - We show the supporting hyperplane inequality:

   $$\begin{aligned} g \in \partial f(x) &\iff \forall y : f(y) \geq f(x) + \langle g, y - x\rangle \\ &\iff \langle (g, -1), (x, f(x))\rangle \geq \sup_y \langle (g, -1), (y, f(y))\rangle = \sup_{(y,t)\in \mathrm{epi}(f)} \langle (g, -1), (y, t)\rangle \end{aligned}$$

   where the first step was by definition, and in the last step we used $(y, t) \in \mathrm{epi}(f) \iff f(y) \leq t$.

   - We claim $L \subseteq \{y \mid \langle g, y\rangle \leq \langle g, x\rangle\} =: H$, which is clearly a supporting hyperplane at x. To show this, assume $y \notin H$ so $\langle g, y\rangle > \langle g, x\rangle$. Then by definition of subgradient we have

   $$f(y) - f(x) \geq \langle g, y - x\rangle > 0,$$

   i.e. $y \notin L$ the sub-level set.

4. Affine and quadratic functions are the most basic convex functions. We will prove some properties about them:

   - Show $h : \mathbb{R}^n \to \mathbb{R}$ is convex *and* concave (i.e. h is convex and $-h$ is convex) iff h is an affine function, i.e. $h(x) = \langle a, x\rangle + b$ for $a \in \mathbb{R}^n, b \in \mathbb{R}$.
   - Let $\tilde{q}(x) := \langle x, Qx\rangle + \langle a, x\rangle + b$ for symmetric matrix $Q \in \mathbb{R}^{n\times n}, a \in \mathbb{R}^n, b \in \mathbb{R}$. Show $\tilde{q}$ is convex iff $q(x) := \langle x, Qx\rangle$ is convex. (Hint: use first part)

- Show $q(x) = \langle x, Qx \rangle$ is convex iff

$$\forall v \in \mathbb{R}^n : \quad \langle v, Qv \rangle \geq 0,$$

and similarly show it is strictly convex iff the inequality is strict for any $v \neq 0$. You can use any of the equivalent characterizations of convexity.
(These conditions are known as positive-semi-definiteness and positive-definiteness, and are denoted $Q \succeq 0, Q \succ 0$.)

- Find the optimizer and optimum value of strictly convex quadratic

$$\tilde{q}(x) := \langle x, Qx \rangle + \langle a, x \rangle + b.$$

Solution:

- Clearly affine h is convex and concave, so we focus on the converse. By convexity and concavity, respectively, we have

$$\lambda h(x) + (1-\lambda)h(y) \leq h(\lambda x + (1-\lambda)y) \leq \lambda h(x) + (1-\lambda)h(y),$$

so in fact we must have equality for all $x, y \in \mathbb{R}^n, \lambda \in [0,1]$. We can use this to explicitly compute the affine function:

$$h(x) = h(0) + \sum_{i=1}^{n} (h(e_i) - h(0))x_i,$$

i.e. $a_i = h(e_i) - h(0), b = h(0)$.

- We use that sums of convex functions are convex, and

$$\tilde{q}(x) = q(x) + h(x), \quad q(x) = \tilde{q}(x) - h(x)$$

where $h(x) := \langle a, x \rangle + b$ is an affine function, so convex and concave. So $\tilde{q}$ is convex iff q is.

- We first note that

$$\begin{aligned}
&\lambda xx^T + (1-\lambda)yy^T - (\lambda x + (1-\lambda)y)(\lambda x + (1-\lambda)y)^T \\
&\qquad = (\lambda - \lambda^2)xx^T + ((1-\lambda) - (1-\lambda)^2)yy^T - \lambda(1-\lambda)(xy^T + yx^T) \\
&\qquad = \lambda(1-\lambda)[xx^T + yy^T - xy^T - yx^T] = \lambda(1-\lambda)(x-y)(x-y)^T.
\end{aligned}$$

Therefore we can rewrite

$$\begin{aligned}
&\lambda q(x) + (1-\lambda)q(y) - q(\lambda x + (1-\lambda)y) \\
&\qquad = \langle Q, \lambda xx^T + (1-\lambda)yy^T - (\lambda x + (1-\lambda)y)(\lambda x + (1-\lambda)y)^T \rangle \\
&\qquad = \lambda(1-\lambda)\langle Q, (x-y)(x-y)^T \rangle.
\end{aligned}$$

Now q is convex iff the above is non-negative for all $x, y \in \mathbb{R}^n, \lambda \in [0,1]$, which is equivalent to the condition $\langle v, Qv \rangle \geq 0$ for $v = x - y$.

- We compute the gradient and check where it vanishes:

$$0 = \nabla \tilde{q}(x^*) = 2Qx + a \quad \iff \quad x^* = -\frac{Q^{-1}a}{2}$$

where we used that $\tilde{q}$ is strictly convex so Q is invertible. Substituting gives

$$\tilde{q}(x^*) = \frac{\langle Q^{-1}a, Q(Q^{-1}a)\rangle}{4} - \frac{\langle a, Q^{-1}a\rangle}{2} + b = -\frac{\langle a, Q^{-1}a\rangle}{4} + b.$$

5. An Ellipsoid is an affine image of the Euclidean ball

$$\mathcal{E} = c + AB_2^n \quad \text{where} \quad B_2^n := \{x \in \mathbb{R}^n \mid \|x\|_2 \leq 1\},$$

for some $c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ invertible.

- Let $q(x) := \langle x, Qx\rangle + \langle d, x\rangle + e$ be *strictly convex*. Show any sub-level set

$$L_t := \{x \in \mathbb{R}^n \mid q(x) \leq t\}$$

is either empty or an Ellipsoid (i.e. find c, A such that $L_t = c + AB_2$).

- Conversely, given Ellipsoid $\mathcal{E} = c + AB_2$ as above, find convex quadratic $q(x) := \langle x, Qx\rangle + \langle d, x\rangle + e$ such that

$$\mathcal{E} = \{x \in \mathbb{R}^n \mid q(x) \leq 1\}.$$

Solution:

- By the last part of the previous question, we can compute the optimizer

$$x^* := \arg\min_x q(x) = -\frac{Q^{-1}d}{2} \quad \text{with} \quad q(x^*) = e - \frac{\langle d, Q^{-1}d\rangle}{4}.$$

so clearly L_t is empty for $t < q(x^*)$. We can rewrite q in terms of the optimizer

$$q(x) = \langle (x - x^*), Q(x - x^*)\rangle + q(x^*),$$

as can be verified directly. Therefore, for any $t \geq q(x^*)$ we have

$$\begin{aligned} x \in L_t &\iff \langle (x - x^*), Q(x - x^*)\rangle \leq t - q(x^*) \\ &\iff \|Q^{1/2}(x - x^*)\|_2 \leq \sqrt{t - q(x^*)} \\ &\iff x - x^* \in \sqrt{t - q(x^*)}\, Q^{-1/2} B_2^n, \end{aligned}$$

where in the last step we used that Q is invertible. This gives the required form for the Ellipsoid.

- We reverse the sequence of equivalences

$$x \in \mathcal{E} = c + AB_2^n \iff x - c \in AB_2^n \iff A^{-1}(x-c) \in B_2^n$$
$$\iff 1 \geq \|A^{-1}(x-c)\|_2^2 = \langle (x-c), (AA^T)^{-1}(x-c) \rangle,$$

so $q(x) := \langle (x-c), (AA^T)^{-1}(x-c) \rangle$ is the required quadratic.

6. Recall the GLS oracle model for convex sets. For convex set $K \subseteq \mathbb{R}^n$, the *polar dual* is

$$K^\circ := \{w \in \mathbb{R}^n \mid \sup_{x \in K} \langle w, x \rangle \leq 1\}.$$

- Relate VAL and VIOL oracles for K to MEM and SEP oracles for K°;
- Combine this with cutting plane methods to show a reduction from SEParation to OPTimization. In more detail, given access to OPT oracle for K, show how to implement the (approximate) SEParation oracle for K.

This may seems counter-intuitive, but there are important settings where OPT is easier to implement than SEP (e.g. when K is the spanning tree polytope of a graph).

Solution:

- We assume for simplicity that 0 is in the interior of K. In this case, all valid halfspaces of K contain the origin, so the value must satisfy $b > 0$. Therefore (w, b) is a VALid hyperplane for K iff

$$\sup_{x \in K} \langle w, x \rangle \leq b \iff \sup_{x \in K} \langle w/b, x \rangle \leq 1 \iff w/b \in K^\circ,$$

i.e. iff MEMbership queried with w/b returns YES. Note that we used $b > 0$ in the second step.

Similarly, assume halfspace (w, b) is VIOLated by $z \in K$ iff z gives a SEParating hyperplane for K°:

$$\sup_{x \in K} \langle w, x \rangle \geq \langle w, z \rangle > b \iff \langle w/b, z \rangle > 1 \geq \sup_{v \in K^\circ} \langle v, z \rangle,$$

where we again used $b > 0$ by assumption, and in the last step we used that $v \in K^\circ \implies \langle v, z \rangle \leq 1$.

- We claim that an OPT oracle for K is equivalent to a SEP oracle for K°, and similarly an OPT oracle for K° is equivalent to a SEP oracle for K. Assuming this claim, the algorithm is as follows: implement a SEP oracle for K° using the

OPT oracle for K; then apply any cutting plane method to get an approximate OPT oracle for K°; then this is equivalent to a SEP oracle for K.

To show the claim, we implement SEP for K° using OPT for K: compute $\sup_{x\in K}\langle w, x\rangle$, and if ≤ 1, output YES; otherwise OPT oracle gives $z \in K$ such that

$$\langle w, z\rangle = \sup_{x\in K}\langle w, x\rangle > 1 \geq \sup_{v\in K^\circ}\langle v, z\rangle,$$

where the strict inequality is because $w \notin K^\circ$ and the last inequality is because $v \in K^\circ \implies \langle v, z\rangle \leq 1$ by definition.

Next, we implement SEP for K using OPT for K°. This follows from the Bi-Polar Theorem $K^{\circ\circ} = K$ for $0 \in K$, so we can reverse the roles of K, K° in the above argument.